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Second Semester, 2026
University of Melbourne

MAST20030: Differential Equations
Revision Tutorial: Week 1

Recall from Linear Algebra

i. *Real vector spaces.* A real vector space is a non-empty set V with two operators: addition and (real) scalar multiplication. These operations are required to satisfy a set of axioms involving the two operations (see your linear algebra notes or https://mathworld.wolfram.com/VectorSpace.html).

ii. *Span.* Let $S = \{\boldsymbol{u}_1, ..., \boldsymbol{u}_k\}$ be a set of vectors from real vector space V. The span of S is the set of all possible linear combinations of the elements of S, i.e.,

$$\text{span}(S) = \left\{ \alpha_1 \boldsymbol{u}_1 + \cdots + \alpha_k \boldsymbol{u}_k \,\middle|\, \alpha_1, ...\alpha_k \in \mathbb{R} \right\}.$$

span(S) is a subspace of V, i.e., span(S) is itself a real vector space.

iii. *Linear independence.* The set of vectors $\{\boldsymbol{u}_1, ..., \boldsymbol{u}_k\}$ is linearly independent if and only if the only solution to $\alpha_1 \boldsymbol{u}_1 + \cdots + \alpha_k \boldsymbol{u}_k = 0$ is $\alpha_1 = 0, ..., \alpha_k = 0$.

iv. *Bases and dimension.* A basis for a vector space V is a set of vectors in V which (1) span V and (2) are linearly independent. The dimension of V is the number of vectors in a basis for V.

Question 1 (Vectors)

(a) Is the set $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\}$ linearly independent?

(b) Show that $\mathbb{R}^2 = \text{span}\left(\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\} \right)$. Using the definition in (iv), what is the dimension of $\mathbb{R}^2$?

Question 2 (Functions)

Consider the functions $f(x) = \sin x$, $g(x) = \cos x$ and $h(x) = 2\sin x$ for $x \in \mathbb{R}$, and assume that f, g and h are elements of some vector space, V, which we leave undefined.

(a) Define span$(\{f, g, h\})$.

(b) Is the set $\{f, g, h\}$ linearly independent?

(c) Is the set $\{f, g\}$ linearly independent?

(d) Find a basis for span$(\{f, g, h\})$ and state its dimension.

Question 3 (Calc 2 revision)

By trialling exponential solutions of the form $e^{\lambda t}$, find the solution to

$$x''(t) + 3x'(t) + 2x(t) = 0 \qquad \text{where} \qquad x(0) = x_0 \qquad \text{and} \qquad x'(0) = v_0 \,.$$

The real-world modelling context is in Case Study 1 problem 2, which is available on the LMS

Question 4 (Eigenvalues and eigenvectors)

Let $\boldsymbol{B}$ be a real $n \times n$ matrix, i.e., a linear transformation. An eigenvector of $\boldsymbol{B}$, is a vector that when acted on by the linear transformation, is mapped to a scalar multiple of itself. Specifically, λ is an eigenvalue of $\boldsymbol{A}$ with associated eigenvector $\boldsymbol{v}$ iff

$$\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v} \ .$$

The eigenvalues are the solutions to the characteristic equation $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$ where $\boldsymbol{I}$ is the $n \times n$ identity matrix. The eigenvectors are then found by finding all linearly independent solutions of $(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{v} = \boldsymbol{0}$, i.e., find a basis for the solution space of $(\boldsymbol{A} - \lambda \boldsymbol{I})$.

(a) Find the eigenvalues and eigenvectors of

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \ . \tag{1}$$

(b) Are the eigenvectors you found in (a) linearly independent?

(c) Do the eigenvectors you found in (b) form a basis for $\mathbb{R}^2$?

Question 5 (Solution Space)

Let $\boldsymbol{A}$ be a real $m \times n$ matrix. $\boldsymbol{A}$ is therefore a linear transformation that maps vectors in $\mathbb{R}^n$ to vectors in $\mathbb{R}^m$, i.e., $\boldsymbol{A} : \mathbb{R}^n \to \mathbb{R}^m$. The solution space of $\boldsymbol{A}$ is the set of all vectors, $\boldsymbol{x}$, such that $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$, i.e., the set

$$\left\{ \boldsymbol{x} \,\middle|\, \boldsymbol{A}\boldsymbol{x} = \boldsymbol{0} \right\} \ .$$

The solution space is also called the nullspace or kernel. The solution space of a linear transformation is also a vector space and so it has bases and dimension just like any other vector space.

Consider the matrix and vector

$$\boldsymbol{A} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \ .$$

(a) Find a basis for the solution space of $\boldsymbol{A}$ and state its dimension.

(b) Find a particular solution (i.e., a single solution) for $\boldsymbol{A}\boldsymbol{y}_p = \boldsymbol{b}$.

(c) Using (a) and (b), construct the general solution to $\boldsymbol{A}\boldsymbol{y} = \boldsymbol{b}$.

Question 6 (Extension question: Symmetric matrices and inner products)

Let $\boldsymbol{A}$ be an $n \times n$ real symmetric matrix, with an associated inner product space given by the dot product. $\boldsymbol{A}$ therefore has n orthogonal eigenvectors and as such these eigenvectors form a basis for $\mathbb{R}^n$.

(a) Find the eigenvalues and eigenvectors of

$$\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \ . \tag{2}$$

(b) Verify that the eigenvectors you found in (a) are orthogonal with respect to the inner (dot) product.

(c) Write an arbitrary vector from $\mathbb{R}^2$ as a linear combination of the elements of your basis from (a), and determine a formula for the coefficients. Hint: use the fact that the eigenvectors of $\boldsymbol{A}$ are orthogonal.